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- 1-Number of Zeros in a Given ...
- 2-Majority Element
- 3-Finding Floor Value
- 4-Two Elements sum to x
- 5-Implementation of Quick Sort
- Dynamic Programming
  - 1-DP-Playing with Numbers
  - 2-DP-Playing with chessboard
  - 3-DP-Longest Common Subse...
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- Competitive Programming
  - 1-Finding Duplicates-O(n^2) Ti...
  - 2-Finding Duplicates-O(n) Tim...
  - 3-Print Intersection of 2 sorte...
  - 4-Print Intersection of 2 sorte...
  - 5-Pair with Difference-O(n^2)T...

RAJALAKSHMI ENGINEERING COLLEGE

Sonam Acharjya 251801322-2025-AIDS S2

Dashboard My courses

CS23331-DAA-2025-CSE / 6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

6-Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

Started on	Friday, 2 October 2026, 11:00 PM
State	Finished
Completed on	Friday, 2 October 2026, 11:08 PM
Time taken	8 mins 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    scanf("%d", &k);

    int found = 0;

    // Brute force nested loops to check every pair (i != j)
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            // Ensure indices are different and the difference equals k
            if (i != j && (arr[j] - arr[i] == k)) {
                found = 1;
                break; // Break the inner loop
            }
        }
        if (found == 1) {
            break; // Break the outer loop early since we found a valid pair
        }
    }

    // Output 1 if a pair exists, otherwise 0
    printf("%d\n", found);

    return 0;
}
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Finish review

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